TECHNICAL CASE STUDY / EPIGRAL / QNAP NAS & DATA MIGRATION

A new storage foundation. A controlled data move.

EPIGRAL Limited

Dahej | IT server room & QC department

QNAP NAS installation and a detailed legacy-data migration design: storage pools, network interfaces, share permissions, staged transfer, cutover and recovery checks.

2 QNAP NAS UNITS

Rackmount + two-bay platform

6 x 4 TB DRIVES

Capacity and backup scope review

Installation orders dated 15 September 2026.
Migration method is a design; completion records were not supplied.



Scope baseline | Storage, connectivity and implementation

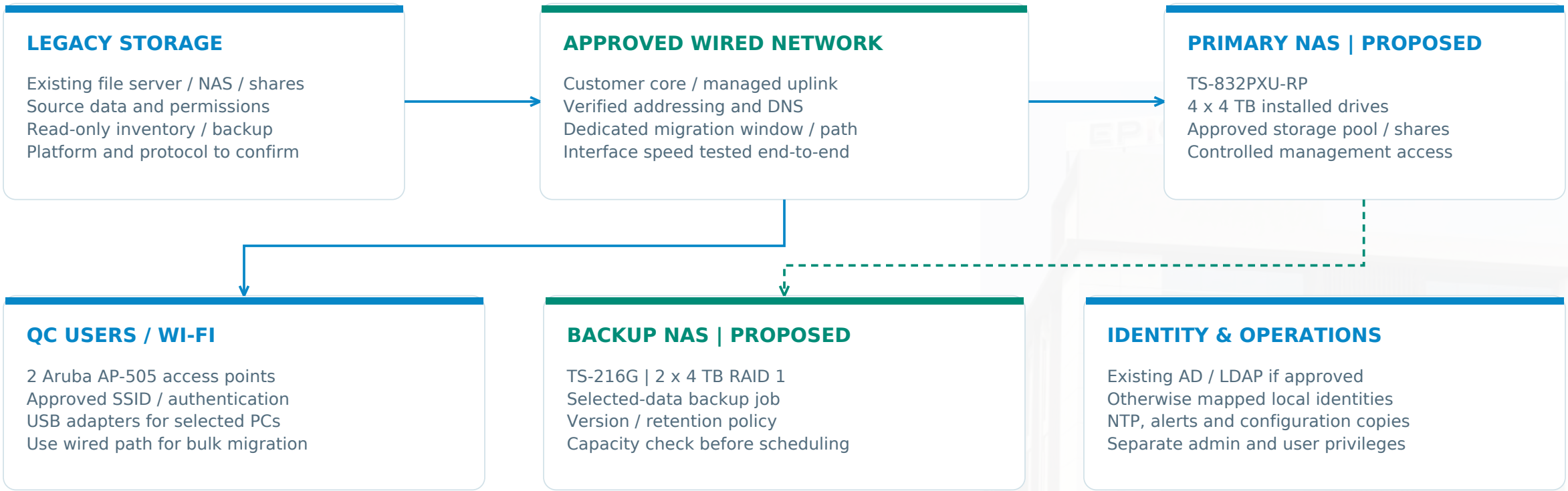
Workstream	Documented supply / service	Design use
Rackmount NAS	1 TS-832PXU-RP: 8 bays, 2U, quad-core 1.7 GHz, 4 GB RAM, redundant PSU; 2 x 10GbE SFP+ and 2 x 2.5GbE; rail kit.	Four ordered drives; primary shared storage role proposed.
Tower NAS	1 TS-216G: 2 bays, 4 GB RAM, 1GbE and 2.5GbE interfaces; 2 x 4 TB drives.	RAID 1 and selected-data backup role proposed; location / role to confirm.
Network & QC	2 Aruba AP-505 APs / mounts, 20 nano USB Wi-Fi adapters, TP-Link LS108GP line, 6U rack, 200 m Cat6 and 100 m conduit.	QC access network; verify switch model, AP power and rack placement.
Implementation orders	4510208051: hardware; 4506131336: IT server-room storage installation; 4506131333: QC storage, AP, switch, rack and cabling services.	Service window: 25 September - 31 October 2026. Legacy migration added from D-TECH’s requested scope.

READINESS GATE

Confirm the old storage platform, live data size, application dependencies, users / groups, current permissions and backup status before sizing the migration. Do not initialise old disks or treat the PO as proof of successful migration.

03 Architecture | 04 Capacity & backup sizing | 05 Commissioning | 06 Legacy migration | 07 Cutover & rollback | 08 Operations & handover.

Architecture | Separate access, migration and recovery paths



10GbE / 2.5GbE NAS ports do not guarantee those speeds through a Gigabit switch. Managed switching and high-speed optics / uplinks are dependencies, not confirmed supply.

NOMINAL DECIMAL CAPACITY / PROPOSED RAID / BACKUP SIZING

Storage design | Capacity must fit the protection policy

TS-832PXU-RP | FOUR POPULATED BAYS

4 x 4 TB = 16 TB raw
Proposed RAID 6: (4 - 2) x 4 TB = 8 TB nominal
Two-disk fault tolerance; no hot spare in this layout
4 empty bays are not usable capacity
Usable space reduces with formatting / reserves.

TS-216G | TWO POPULATED BAYS

2 x 4 TB = 8 TB raw
Proposed RAID 1: 4 TB nominal usable
One-disk fault tolerance
Backup data + versions + headroom must fit
RAID / snapshots do not replace separate backup.

Sizing check	Decision before initialisation
BACKUP CAPACITY	A 4 TB target cannot hold a full 8 TB primary dataset. Select an approved critical-data subset or expand the backup design; include versions and growth allowance.
RAID / VOLUME TYPE	RAID 6 is proposed, not confirmed installed. Approve resilience, performance and capacity tradeoffs; verify supported QTS build, pool / volume type and snapshot capability.
DATA GROWTH	Inventory current used bytes, expected growth, largest files and small-file count. Set warning thresholds, quotas and snapshot reserve from customer policy, not arbitrary guarantees.

All figures are pre-format decimal estimates. Drive compatibility, firmware, actual storage allocation and customer approval determine the final configuration.

Commissioning | Build the NAS before moving production data

Step	Configuration activity	Acceptance check
01 | RACK & DISKS	Verify serials / drive compatibility; map bay order; mount rails / NAS in a suitable IT rack; separate PSU feeds where available.	Airflow / power / drive health; no source-disk reuse or destructive source action.
02 | INITIALISE	Choose supported firmware, administrator identity and device name; assign approved IP / DNS / NTP; configure restricted management access.	Correct time / routing; named admin access; configuration copy.
03 | POOL & VOLUME	Create approved RAID group / storage pool and volume; allocate headroom and snapshot reserve where supported.	Healthy RAID sync; expected usable capacity and alerts.
04 | IDENTITY & SHARES	Join approved directory or map local users; create shares, quotas and explicit share-level permissions; configure compatible SMB access.	Read / write / deny tests with representative users; no broad-access shortcut.
05 | LAN & WI-FI	Check switch / uplink rate and AP PoE budget. Keep standard MTU unless end-to-end support is verified; configure SSIDs as approved.	Wired throughput pilot, AP reachability and client authentication.
06 | PROTECTION	Configure approved snapshots / backup jobs and notifications; test a recovery to an isolated target before cutover.	Successful job / restore evidence, owners and monitoring schedule.

The LS108GP order line is labelled PoE; verify exact supplied model / power budget. Do not assume unmanaged switching can enforce VLAN policy or provide a 10GbE backbone.

Legacy migration | Inventory, pilot, seed and reconcile

Stage	Micro-level procedure	Evidence / decision
INVENTORY	Record source paths, used bytes, file counts, timestamps, owners / groups, ACLs, open files, long paths and dependent applications. Take a recoverable source backup.	Approved source-to-target share map; migration exclusions and owner.
PILOT	Copy a representative folder with large / small files and restricted access. Choose tooling for the source: compatible HBS NAS jobs or an approved file-copy tool.	Test contents, metadata and access before scaling the job.
INITIAL SEED	Copy while source remains authoritative. Use a dedicated service identity; keep logs, retry limits and exception list. Do not enable deletion / mirror purging by default.	Transferred bytes / counts, skipped files and errors recorded.
IDENTITY / ACL	Match directory identities or explicit local mappings. Verify HBS ACL support for the selected path; recreate destination share permissions separately.	Authorised read / write and unauthorised denial tests.
VALIDATE	Compare folder / file counts, bytes and timestamps; perform agreed content / checksum checks for critical files; reconcile renamed, locked or unsupported files.	All critical exceptions resolved or formally accepted.
FINAL DELTA	Schedule write freeze; close application sessions as agreed; copy changed files and repeat validation. Protect both source and target through the cutover window.	Verified delta and customer go / no-go approval.

QNAP HBS can replicate supported ACLs; it does not automatically recreate share-level permissions. HBS options depend on installed version and source / destination type.

CHANGE WINDOW / CUSTOMER ACCEPTANCE / RECOVERY DECISION

Cutover & rollback | One authoritative copy at a time

01 | FREEZE & DELTA

Agree window, stop source writes, close sessions, run final delta and resolve copy exceptions.

02 | GO / NO-GO

Check RAID / share health, data reconciliation, ACL tests and successful backup / restore evidence.

03 | SWITCH ACCESS

Change approved mapped drives / share paths or application references. Avoid concurrent writing to both copies.

04 | USER VALIDATION

Confirm critical folders, read / write / denial, application links and representative file content with owners.

05 | MONITOR

Watch errors, storage capacity, client access and backup results. Keep the old source protected for agreed retention.

06 | ACCEPT OR ROLL BACK

If rollback is required, stop new-target writes and reconcile target changes before directing users back to source.

Live databases / instrument software require application-owner migration methods and consistent exports or backups; copying open data files is not an application-consistent migration.

Operations & handover | Make recovery repeatable

EPIGRAL | CUSTOMER DETAILS

EPIGRAL Limited, Dahej Works
CH/1 and CH/2, GIDC Industrial Estate
Taluka Vagra, Bharuch, Gujarat 392130.
IT server-room and QC implementation orders dated 15 September 2026.

TECHNICAL HANDOVER PACK

NAS / disk serials, bay map, RAID / pool layout, interface plan, share / ACL register, migration logs, exceptions, backup / restore records, alert owners, configuration copies and rollback instructions.

Operating check	Record and ownership
STORAGE / ACCESS	Monitor disk / RAID health, pool capacity, snapshot use and failed logins; review permissions and alerts with the customer IT owner.
BACKUP / RECOVERY	Review selected-data jobs, failures, versions and target free space; schedule restore tests. Agree RPO / RTO from business needs; no achieved recovery figures supplied.

D-TECH Customer Care | support.dtechindia.com | +91 99989 03042

Basis: printed orders 4510208051, 4506131336 and 4506131333; user-requested legacy migration. QNAP QTS / HBS references support the design method.

Reference links: QTS Storage & Snapshots | HBS sync job options | ACL versus share permissions. Cover uses the supplied EPIGRAL building photo.